

CSIS 4440(Sahan Nonis 300389470)

F25_4440_G#7

Progress Report

Date	Hours	Description of Work Done
15th Sep 2025	0.5	Group formed and agreed on individual roles.
6th Oct 2025	1.0	Conducted research on tools and applications for analysis.
9th Oct 2025	0.5	Met with professor to discuss project scope and confirm selected tools & apps.
22nd Oct 2025	1.0	Researched and set up Drozer and JADX; drafted assessment plan for Trust Wallet & Spotify.
1st Nov 2025	3	Setup drozer and made a successful connection to the emulator. encountered error connecting it to the emulator and getting the right ip address
6th Nov 2025	1	Tested Drozer connectivity to an Android emulator, learned how to start the Drozer agent, and explored basic commands (run app.package.list, run app.package.info).
8th Nov 2025	1.5	Installed JADX-GUI and JADX-CLI, explored interface layout, and learned preferences for decompilation accuracy.
9th Nov 2025	2	Practiced decompiling a sample APK using JADX, inspected file tree, learned how to navigate Java source, resources, and smali.
9th Nov 2025	1.5	Researched Drozer modules for attack surface mapping (scanner.activity, scanner.provider, scanner.broadcast).
10th Nov 2025	1	read articles on common Java decompilation issues.
13th Nov 2025	1.0	Practiced identifying exported components using Drozer; tested simple exploitation of exported activities using Drozer commands.

15th Nov 2025	2.0	Prepared notes for progress report.
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Description of Work Done

So far, I have spent time learning and experimenting with the two forensic/security tools I selected: Drozer and JADX. I began by setting up both tools, installing dependencies, and ensuring they run smoothly on my analysis environment. With Drozer, I learned how the agent communicates with the framework, explored key modules, scanned for exported components, and tested basic exploitation techniques such as querying insecure content providers and launching exposed activities. With JADX, I practiced decompiling APKs, navigating Java source, examining manifest files, and identifying potential insecure code patterns.

The main issues I encountered involved connecting Drozer to the Android emulator reliably, which required adjusting ADB port forwards. I also encountered minor limitations in JADX when dealing with obfuscated APKs, but I researched how to use search and smali navigation to get around these issues.

For next steps, I plan to select the two novel applications I will analyze for the final project, begin full static + dynamic forensic analysis, and integrate both tools into a structured workflow. I will also continue documenting findings, capturing screenshots, and preparing detailed methodology for the final report.

AI Use Section:

AI Tool Name	Version, Account Type	Specific feature For which the AI tool was used
ChatGPT	GPT 5.0, Plus	To polish report and to debug errors

Value Addition:
I used AI only for clarification and structuring my work logs. All hands-on testing, setup, tool usage, and analysis of Drozer and JADX were performed by me directly on my environment.

Appendix:
Can you explain how Drozer modules work and which ones are useful for attack surface mapping?"

Show me how to decompile an APK using JADX and what security findings I should look for.

Help me outline a daily work log for learning Drozer and JADX for my project.